MUKESH ETHIRAJ

College Park, MD | +1 (917) 601-5456 | mukeshethiraj0@gmail.com | linkedin.com/in/mukesh-ethiraj/

PROFESSIONAL SUMMARY

Data analyst with experience in hospitality, logistics, and retail analytics, building dashboards, KPI reporting, and automated reports with SQL, Python, Power BI, Tableau, Excel, and Snowflake. Also builds data warehouses and pipelines with dbt, PostgreSQL, and Kafka, with strengths in data validation, reconciliation, and root-cause analysis.

TECHNICAL SKILLS

BI & Visualization: Power BI, Tableau, Excel, Alteryx, Oracle Hospitality Cloud, Matplotlib, Seaborn
Data & Programming: SQL, Python (Pandas, NumPy), Snowflake, Azure SQL, PostgreSQL, dbt, Kafka, Docker, ETL/ELT
Analysis & Reporting: Data Visualization, KPI Reporting, EDA, Trend Analysis, Root-Cause Analysis, Ad-hoc Analysis
Data Quality: Data Cleaning, Data Validation, Data Reconciliation, Data Modeling, SQL Optimization, Report Automation

PROFESSIONAL EXPERIENCE

Operational Data Analyst (Intern) | Sonesta International Hotels | Washington, DC — Jan 2026 – Jun 2026

- Built and maintained 15+ recurring SQL, Excel, and Power BI reports tracking repeat-stay rate, campaign response, booking channels, and guest lifetime value, giving marketing and loyalty teams weekly and monthly performance insights.
- Analyzed CRM, guest-stay, and PMS/CRS booking data in Oracle Hospitality Cloud across 30+ hotel properties, using SQL to measure revenue metrics including occupancy, ADR, RevPAR, booking pace, and channel mix.
- Ran root-cause analysis on booking and revenue anomalies, validating and reconciling source data to improve accuracy.
- Partnered with BI and marketing teams to automate 5+ reports, standardizing reports and cutting ad-hoc requests by ~25%.

Marketing Analytics Intern (Logistics Data) | Wizcom Corporation | Princeton, NJ — Jun 2024 – Dec 2024

- Built Tableau dashboards on Snowflake visualizing 50K+ monthly shipment records across European markets, comparing carrier efficiency and delivery outcomes by carrier, region, and reporting period.
- Defined delivery-success, transit-time, and failure-rate KPIs to standardize how carrier performance was measured.
- Automated cleaning of third-party shipment data with Python (Pandas) and dbt, reducing manual reporting effort by ~35%.
- Optimized Snowflake SQL joins and queries, cutting Tableau dashboard load time by 80% (from 20s to 4s).

Data Analyst (Part-time) | GBS Systems & Services — Dec 2022 – Aug 2023

- Built a Power BI dashboard on Azure SQL with REST API integrations to monitor order delays and warehouse metrics.
- Automated recurring operational reports and ETL workflows to consolidate source data and standardize KPI calculations.
- Cleaned and validated reporting datasets with SQL and Python, resolving inconsistencies through reconciliation checks.

PROJECTS

TT&T Support AI – Agentic Telecom Support Assistant | Python, FastAPI, CrewAI, MCP, PostgreSQL, pgvector, React

- Built a full-stack AI support assistant that explains bill changes, answers policy questions, checks outages, and creates support tickets, routing each request to a CrewAI agent that acts only through restricted MCP tools.
- Designed a hybrid RAG pipeline combining pgvector semantic search, PostgreSQL full-text search, Reciprocal Rank Fusion, and cross-encoder reranking with a local Llama 3.2 model, reaching 92.6% hit rate@5 and 0.80 MRR.
- Enforced session-scoped authorization to block cross-account requests, added idempotent ticket creation, and returned source document IDs with every grounded answer.

Bank Regulatory Reporting Warehouse | Python, PostgreSQL, dbt, Docker, Metabase

- Built a raw → staging → warehouse → marts pipeline in dbt and PostgreSQL with SCD Type 1/2/3 customer history and row-level lineage back to source files.
- Caught a cross-batch duplicate bug with automated reconciliation tests (414 accounts off by $186.9K); fixed it so all 520K+ transactions match official balances to the cent.
- Benchmarked indexing vs. partitioning on 6M rows; partitioning kept 200x faster lookups while reloading 3.5x faster than indexed tables.

Near-Real-Time Fraud Detection Pipeline | PostgreSQL, Debezium, Kafka, Python, GitLab CI

- Streamed PostgreSQL changes via Debezium CDC and Kafka to a Python rules engine (velocity, amount anomaly, impossible travel), catching 553/553 planted fraud cases.
- Delivered 0.71s average detection latency with effectively exactly-once alerts, backed by 12 unit tests in GitLab CI.

Image Classification with Mini-ResNet | Python, PyTorch

- Developed a 4.9M-parameter ResNet for CIFAR-10, adapting model code to support custom block depths under a 5M limit.
- Ran controlled experiments on learning-rate schedulers, dropout, weight decay, data augmentation, and kernel size; the selected model reached 92.1% test accuracy.

EDUCATION & CERTIFICATIONS

New York University | M.S. Computer Engineering & Finance | Graduate Merit Scholarship — May 2025
Anna University | B.E. Computer Science — Jul 2023
Training: Microsoft Power BI Data Analyst | AWS Cloud Technical Essentials | Tableau Specialization (UC Davis)